# Unified pipeline: implemented boundaries and current evidence

One raw-data registry, one participant fold plan, shared train/apply ComBat.
Ridge and GINE use identical paired observations for DWI, fMRI and fusion.

Implemented

- Training-only edge QC, harmonization and normalization
- Inner model / sparsity selection and independent GINE stopping
- Inner-OOF reference calibration; global and cohort-aware cBAG
- Original outer checkpoints, private embeddings, gradient contributions
- Runtime audits, resumable tasks and source snapshots

Run classification: development evaluation — external confirmation still required

# Outer / inner / stopping boundaries

Raw DWI log1p / signed fMRI Fisher-z → fixed participant outer split

OUTER TRAINING CN
Inner train / validation split
↓ GINE only: split inner training into fit / stopping
Fit preprocessing on fit only; choose epoch on stopping
↓ Refit preprocessing/model on all inner training
Evaluate inner validation; choose configuration
↓ Calibrate using inner OOF predictions
Refit outer model for inner-selected epoch budget

OUTER HELD-OUT PARTICIPANTS
Frozen preprocessing → raw age prediction
↓ Compute BAG using actual age
↓ Apply frozen training-only calibration

No test age in feature transformations.
No test fold in hyperparameters / sparsity / epochs.
No clinical, genetic or metabolic predictor features.

# Harmonization decision follows age-preservation evidence

Primary development candidate: training-age mean-only ComBat
Held-out transformation: x_harmonized = x − training residual SD × batch offset

Training age helps separate age and acquisition offsets; held-out age is never required.
No-ComBat remains a prespecified control. Batch-only location/scale remains a diagnostic sensitivity option.
Prior modality-wide MAE: DWI none 6.02 / mean-only 6.11; fMRI none 7.43 / mean-only 7.49.
Those historical development comparisons are not the new matched six-model benchmark.

# Conservative population: unresolved pairs are not averaged

| Population | Scans | Participants | CN scans |
|---|---|---|---|
| DWI | 919 | 670 | 723 |
| fMRI | 575 | 447 | 441 |
| paired | 318 | 252 | 237 |

Pairing: 318 retained; 250 age_conflict; 7 insufficient_paired_CN_support.
Same-key / age / sex / diagnosis agreement is checked; acquisition/clinical dates still require verification.

# GINE comparisons and scientific outputs

Capacity: 1 × 16, 2 × 32, current 3 × 64; 2 × 32 residual ablation.
Graph sparsity is selected inside inner validation. Signed fMRI edges are retained; fusion uses modality-union topology and concatenated node profiles.

Learning curves: evaluation-mode training / stopping loss and MAE; selected epochs recorded.
Separate runs vary training participant fractions with fixed outer test folds.

Bias correction: inspect global and cohort-aware OOF cBAG versus age within every cohort.

Interpretation: original outer checkpoints, fold-specific embeddings and gradient-times-input.
Anatomical naming requires verified ROI order.

# Results: completion status is part of the result

| Mode | Input | Model | Completed folds | Mean fold MAE |
|---|---|---|---|---|
| age_mean | DWI | ridge | 5 | 6.239 |
| age_mean | fMRI | ridge | 5 | 8.065 |
| age_mean | fusion | ridge | 5 | 6.461 |
| none | DWI | ridge | 5 | 6.364 |
| none | fMRI | ridge | 5 | 8.035 |
| none | fusion | ridge | 5 | 6.476 |

Status: complete. Mean-fold values are descriptive; use the report for pooled OOF metrics and cohort breakdowns.

# Remaining data validation and clinical analysis gates

Verify HABS diagnosis provenance, ADRC visit links/protocols, age discrepancies and ROI order.

Clinical inventory: CN/MCI/AD, APOE / APOE4, sex, T2D, HbA1c and continuous age. Source codes, units and clinical timing must be verified before testing.

Clinical analyses: group distributions and complete-case multivariable associations with cohort, age and sex; participant-cluster uncertainty; explicit exploratory multiplicity correction.

Embeddings: train-fit PCA per outer model, held-out overlays; never pool unaligned fold coordinates.

No claim of fully removed scanner effects, preserved biology, clinical biomarkers or unseen-site generalization.